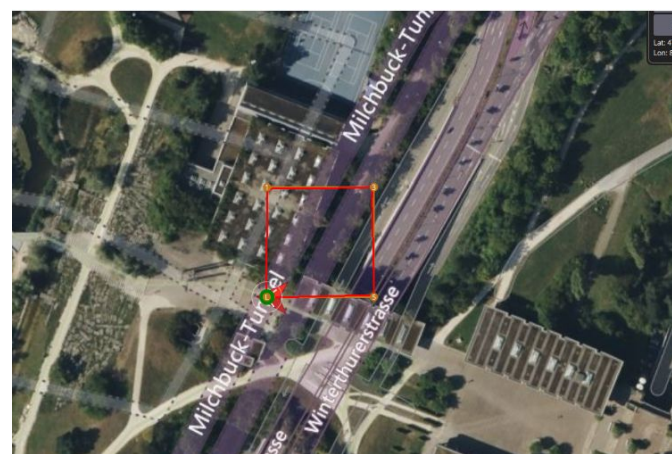


AeroShield

Work Done

- Coded out simple missions
- Created human readable logger
- Tried to capture binary logs for future injections
- Tried to set up the drone with qgc + px4



```
# drone_trace_20251009T100137Z_sqare_mission.txt
1 [2025-10-09T19:01:37.121Z] LOGGER start mission:sqare_mission
2 [2025-10-09T19:01:37.129Z] PX4 VERSION 1.16.0 (967Webb0x0)
3 [2025-10-09T19:01:37.161Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
4 [2025-10-09T19:01:37.329Z] FLIGHT_MODE HOLD
5 [2025-10-09T19:01:37.338Z] HEALTH gps_ok=false home_ok=true local_ok=true
6 [2025-10-09T19:01:37.351Z] BATTERY 10000,0% 16.20V
7 [2025-10-09T19:01:37.364Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
8 [2025-10-09T19:01:37.568Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
9 [2025-10-09T19:01:37.768Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
10 [2025-10-09T19:01:37.978Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
11 [2025-10-09T19:01:38.172Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
12 [2025-10-09T19:01:38.384Z] BATTERY 10000,0% 16.20V
13 [2025-10-09T19:01:38.357Z] FLIGHT_MODE HOLD
14 [2025-10-09T19:01:38.358Z] HEALTH gps_ok=true home_ok=true local_ok=true
15 [2025-10-09T19:01:38.374Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
16 [2025-10-09T19:01:38.578Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
17 [2025-10-09T19:01:38.788Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
18 [2025-10-09T19:01:38.947Z] MISSION_PROGRESS current=0 total=5
19 [2025-10-09T19:01:38.982Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
20 [2025-10-09T19:01:39.184Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
21 [2025-10-09T19:01:39.338Z] BATTERY 10000,0% 16.20V
22 [2025-10-09T19:01:39.347Z] FLIGHT_MODE MISSION
23 [2025-10-09T19:01:39.388Z] HEALTH gps_ok=true home_ok=true local_ok=true
24 [2025-10-09T19:01:39.398Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
25 [2025-10-09T19:01:39.592Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
26 [2025-10-09T19:01:39.794Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
27 [2025-10-09T19:01:39.996Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
28 [2025-10-09T19:01:40.197Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
29 [2025-10-09T19:01:40.338Z] BATTERY 10000,0% 16.20V
30 [2025-10-09T19:01:40.408Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
31 [2025-10-09T19:01:40.413Z] FLIGHT_MODE MISSION
32 [2025-10-09T19:01:40.413Z] HEALTH gps_ok=true home_ok=true local_ok=true
33 [2025-10-09T19:01:40.681Z] POSITION lat=47.3979789 lon=8.5461634 rel_alt_m=0.00
```

Challenges

- Was unable to capture logs due to port and id conflicts from px4
- Learning MAVSDK, QGC, PX4 when connected to an actual drone
 - Retrieving vehicle information and data
 - Setting up QGC when connected to the drone itself
 - Hooking up the drone causes endless, automatic flight mode switching
 - Possibly because we haven't set up an RC connection
 - Figuring out vehicle configuration components

Next Week

- Fully connect drone to qgc+px4
- Finish setup to capture binary files
- Create mission replays based on logs and add injections
- Schedule meeting with Tom for some time next week